Objective-conditioned speedup and FPGA utilization

Speedup over baseline (×, log scale)

LATENCY AREA ADP

	LATENCY	AREA	ADP
Kalman	32×	1.6×	32×
GAN	1.1e+02×	39×	74×
trmm-opt	25×	1×	21×
covariance	3.3×	1×	2.5×
jacobi-2d	9×	9×	8.2×
syr2k	1.8×	1×	1.6×
chstone-aes	1.8×	0.52×	0.62×
chstone-jpeg	1.7×	1.7×	1.6×
rosetta-3d-rendering	7.6×	7.6×	7.6×

LATENCY

	BRAM	DSP	FF	LUT
Kalman	96.8	0.1	1.2	7.8
GAN	18.3	14.4	7.8	12.8
trmm-opt	0.0	1.6	57.1	75.9
covariance	0.0	1.6	10.5	8.7
jacobi-2d	0.0	1.4	4.6	7.8
syr2k	0.0	1.4	4.1	9.8
chstone-aes	2.6	0.0	0.8	5.7
chstone-jpeg	9.9	67.3	10.1	71.7
rosetta-3d-rendering	10.3	0.2	0.7	2.9

AREA

	BRAM	DSP	FF	LUT
Kalman	95.8	0.4	0.6	1.4
GAN	18.3	14.4	7.2	11.3
trmm-opt	0.0	0.8	0.2	0.6
covariance	0.0	0.8	0.3	0.9
jacobi-2d	0.0	1.2	0.6	1.3
syr2k	0.0	1.4	0.4	0.8
chstone-aes	1.8	0.0	0.7	4.7
chstone-jpeg	9.5	45.1	8.0	50.4
rosetta-3d-rendering	10.1	0.2	0.5	1.6

ADP

	BRAM	DSP	FF	LUT
Kalman	96.8	0.1	1.2	7.8
GAN	18.3	14.4	7.1	11.6
trmm-opt	0.0	1.6	17.2	22.3
covariance	0.0	1.0	0.5	2.0
jacobi-2d	0.0	1.0	1.3	2.7
syr2k	0.0	1.4	0.9	2.7
chstone-aes	1.9	0.0	0.7	5.3
chstone-jpeg	9.1	67.4	10.0	71.6
rosetta-3d-rendering	11.2	0.2	0.5	1.9

Per-resource utilization (% of device budget)